



CEH v13 — Module 03: Scanning Networks



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Scanning Networks



Module Overview

This module covers network scanning techniques, including host discovery, port scanning, service enumeration, OS fingerprinting, firewall evasion, and defensive countermeasures. Scanning is the second phase of ethical hacking after reconnaissance.

1. Network Scanning Concepts

1.1 Overview of Network Scanning

Network scanning is the process of systematically probing a network to identify active hosts, open ports, running services, and security configurations.

Objectives of Network Scanning

Live Host Identification

Determining which IP addresses are assigned to active systems

Port State Discovery

Identifying open, closed, and filtered ports on target systems

Service Enumeration

Detecting running services and their versions

Operating System Detection

Fingerprinting the OS and version of target systems

Network Topology Mapping

Understanding network architecture and device placement

Security Policy Analysis

Identifying filtering devices such as firewalls, IDS/IPS, and load balancers



Position in the Attack Lifecycle

Scanning is the **second phase** of ethical hacking, immediately following reconnaissance. It transitions from passive information gathering to active probing of target systems.

1.2 TCP Communication Flags

TCP uses **control flags** in the header to manage connection state and data flow.

TCP Flag Reference

Flag	Meaning	Purpose
SYN	Synchronize	Initiate a connection
ACK	Acknowledgment	Acknowledge received data
FIN	Finish	Graceful connection termination
RST	Reset	Abort connection immediately
PSH	Push	Push buffered data to application
URG	Urgent	Urgent pointer field is significant



Significance for Port Scanning

These flags form the basis of many **port-scanning techniques**. By manipulating flag combinations, attackers can:

- Evade detection by IDS/IPS systems
- Bypass firewall rules
- Fingerprint operating systems
- Determine port states indirectly

1.3 TCP/IP Communication

Understanding TCP/IP communication is essential for effective network scanning.

TCP Three-Way Handshake

Normal connection establishment:

```
Client → Server: SYN
Server → Client: SYN/ACK
Client → Server: ACK
[Connection established]
```

Connection Termination

Graceful Termination:

```
Client → Server: FIN
Server → Client: FIN/ACK
Client → Server: ACK
```

Immediate Termination:

```
Either side: RST
```

ICMP in Host Discovery

ICMP (Internet Control Message Protocol) is commonly used for:

- Host discovery (ping)
- Network diagnostics
- Error reporting
- Path MTU discovery



UDP Scanning Challenge

UDP is stateless and connectionless, making scanning difficult:

- No handshake or session establishment
- Requires indirect methods for scanning
- Port state inferred from **ICMP Port Unreachable** responses
- Open ports often yield no response (timeout)
- Makes UDP scanning slower and less reliable than TCP

2. Scanning Tools

2.1 Nmap



Nmap: The Industry Standard

Nmap (Network Mapper) is the most widely used network scanning tool in the security industry, offering comprehensive capabilities for host discovery, port scanning, and service enumeration.

Core Capabilities

Host Discovery

Identifying live systems using various ping techniques

Port Scanning

Multiple scan types (SYN, Connect, UDP, FIN, NULL, Xmas, etc.)

OS Detection

TCP/IP stack fingerprinting to identify operating systems

Version Detection

Service and application version identification

NSE Scripting

Nmap Scripting Engine for automation and vulnerability detection

Firewall Evasion

Techniques to bypass filtering devices

IPv6 Support

Native IPv6 scanning capabilities

Common Nmap Commands

```
# Basic host discovery
nmap -sn 192.168.1.0/24

# SYN scan (stealth)
nmap -sS 192.168.1.10

# Full port scan with version detection
nmap -p- -sV 192.168.1.10

# OS detection
nmap -O 192.168.1.10

# Aggressive scan with scripts
nmap -A 192.168.1.10

# UDP scan
nmap -sU 192.168.1.10
```

2.2 Hping3

Hping3 is a powerful packet crafting and network testing tool.

Primary Uses

Custom Packet Crafting

Create TCP/UDP/ICMP packets with specific flags and payloads

Advanced Firewall Testing

Test firewall rules and filtering behavior

Idle Scans

Perform stealthy scans using zombie hosts

Traceroute

Custom traceroute with various protocols

TTL Manipulation

Modify Time-To-Live values for fingerprinting and evasion

Hping3 Examples

```
# SYN scan on port 80
hping3 -S -p 80 192.168.1.10

# UDP scan
hping3 -2 -p 53 192.168.1.10

# ICMP flood test
hping3 -1 --flood 192.168.1.10

# Custom TTL
hping3 -S -p 80 -t 5 192.168.1.10
```

2.3 Hping Scan with AI

AI enhances Hping3's capabilities:

AI Enhancements

- **Automating packet modification** based on target responses
- **Identifying anomalies in replies** that indicate filtering or deception
- **Pattern-based port/state inference** from incomplete or ambiguous responses
- **Creating custom payloads** based on target behavior analysis

- **Adaptive timing** to optimize scan speed while avoiding detection
- **Intelligent retry logic** for unreliable network conditions

2.4 Metasploit

Metasploit Framework includes powerful auxiliary scanning modules.

Scanning Capabilities

Auxiliary Scanning Modules

Pre-built scanners for various protocols and services

Protocol-Specific Scanners:

- **SMB** — SMB version detection, share enumeration
- **SSH** — SSH version and algorithm detection
- **FTP** — Anonymous FTP detection, version identification
- **SNMP** — SNMP enumeration and community string testing

Automated Vulnerability Checks

Integrated vulnerability scanning and exploitation

Metasploit Scanner Examples

```
# SMB version detection
use auxiliary/scanner/smb/smb_version

# SSH version scan
use auxiliary/scanner/ssh/ssh_version

# Port scanner
use auxiliary/scanner/portscan/tcp
```

2.5 NetScanTools Pro

NetScanTools Pro is a comprehensive commercial network scanning suite.

Features

DNS Tools

DNS lookup, zone transfer testing, reverse DNS

Ping Sweeps

ICMP-based host discovery across ranges

ARP Scanning

Local network MAC address discovery

Packet Capture

Network traffic analysis and inspection

Banner Grabbing

Service identification through banner collection

Additional Utilities:

- WHOIS lookup
 - Traceroute visualization
 - Port scanner
 - Network connection monitor
-

3. Host Discovery



Primary Goal

Host discovery aims to identify **which IP addresses correspond to live, active hosts** on the target network before conducting more intensive port scanning.

3.1 Host Discovery Techniques

Five Primary Categories

ICMP-Based Discovery

Using ICMP echo, timestamp, and address mask requests

TCP-Based Discovery

Sending TCP packets (SYN, ACK) to common ports

UDP-Based Discovery

Sending UDP packets to elicit responses

ARP-Based Discovery

ARP requests on local networks (Layer 2)

Protocol-Based Discovery

Using IGMP, IP protocol scans, and IP options

3.2 ARP Ping Scan

ARP (Address Resolution Protocol) scanning is highly effective on local networks.

Characteristics

- Works only on **local LAN** (same broadcast domain)
- Cannot be blocked by host firewalls (operates at Layer 2)
- Bypasses ICMP filtering completely
- Most reliable method for local network discovery
- Instant response if host is alive

Nmap ARP Scan

```
# Automatic ARP scan on local network
nmap -sn 192.168.1.0/24

# Explicit ARP ping
nmap -PR 192.168.1.0/24
```

3.3 UDP Ping Scan

UDP ping sends empty or small UDP packets to target hosts.

Response Interpretation

ICMP Port Unreachable Response

Host is alive (port is closed)

No Response

Port is open or filtered (ambiguous)

UDP Service Response

Host is alive and port is open

Nmap UDP Ping

```
nmap -PU 192.168.1.10
```

3.4 ICMP Echo Ping Scan

The **classic ping** using ICMP Echo Request (Type 8).

Characteristics

- Most common host discovery method
- Often blocked by firewalls and host-based filtering
- Simple and fast when allowed
- ICMP Echo Reply (Type 0) indicates host is alive

Usage

```
# Standard ping  
ping 192.168.1.10
```

```
# Nmap ICMP echo ping  
nmap -PE 192.168.1.10
```

3.5 ICMP Echo Ping Sweep

Ping sweep scans multiple hosts in a range for live systems.

Tools

fping

Fast, parallelized ping utility

```
fping -a -g 192.168.1.0/24
```

Angry IP Scanner

GUI-based IP scanner with ping sweep capabilities

Nmap

```
nmap -sn -PE 192.168.1.0/24
```

3.6 ICMP Timestamp Ping Scan

ICMP Timestamp Request (Type 13) queries system time.

Advantages

- Rarely filtered by firewalls
- Alternative when ICMP Echo is blocked
- Can reveal timezone information
- Useful for clock skew analysis

Nmap Command

```
nmap -PP 192.168.1.10
```

3.7 ICMP Address Mask Ping Scan

ICMP Address Mask Request (Type 17) queries subnet mask.

Intelligence Gathered

- Reveals network class and subnet design
- Often overlooked by security teams
- Less commonly filtered than Echo requests
- Can aid in network topology mapping

Nmap Command

```
nmap -PM 192.168.1.10
```

3.8 TCP SYN Ping Scan

TCP SYN ping sends SYN packets to commonly open ports.

Response Analysis

SYN/ACK Response

Host is alive, port is open

RST Response

Host is alive, port is closed

No Response

Port is filtered or host is down

Common Target Ports

- Port 80 (HTTP)
- Port 443 (HTTPS)
- Port 22 (SSH)
- Port 3389 (RDP)

Nmap Command

```
# SYN ping to ports 80 and 443  
nmap -PS80,443 192.168.1.10
```

3.9 TCP ACK Ping Scan

TCP ACK ping sends packets with the ACK flag set.

Purpose

- Used to bypass **stateless firewalls**
- Detects stateful vs. stateless filtering behavior
- RST response indicates host is alive
- Useful when SYN packets are filtered

Nmap Command

```
nmap -PA80,443 192.168.1.10
```

3.10 IP Protocol Ping Scan

IP Protocol Scan sends packets with different IP protocol numbers (1-255).

Characteristics

- Tests which IP protocols are supported
- ICMP Protocol Unreachable indicates host is alive
- Useful for fingerprinting and discovery
- Rarely used but effective when other methods fail

Nmap Command

```
nmap -PO 192.168.1.10
```

3.11 Host Discovery with AI

AI enhances host discovery through intelligent analysis.

AI Capabilities

Automatic Pattern Inference

Learning which discovery methods work best for specific targets

Adaptive Retry Logic

Intelligently adjusting timeouts and retry attempts

Predicting Live Hosts

Inferring host status from partial or ambiguous responses

Reducing False Positives

Machine learning to distinguish real hosts from honeypots

Detecting Stealth Firewall Rules

Identifying filtering patterns through response analysis

Optimizing Scan Order

Prioritizing likely-live hosts based on network patterns

3.12 Ping Sweep Tools

Popular Tools

fping

Fast, scriptable ping utility with range support

Nmap

Comprehensive host discovery with multiple techniques

Angry IP Scanner

User-friendly GUI scanner for Windows/Linux/Mac

SolarWinds IP Scanner

Enterprise-grade commercial IP scanning tool

Masscan

Extremely fast port scanner with ping capabilities

Netdiscover

ARP-based network discovery tool

4. Port and Service Discovery

4.1 Port Scanning Techniques

4.1.1 TCP Connect / Full-Open Scan

Full TCP connection using the complete three-way handshake.



Characteristics

- Uses full 3-way handshake: SYN → SYN/ACK → ACK
- Most accurate port state detection
- Easily detected and logged by target systems
- Does not require root/administrator privileges
- Slower than half-open scans

Nmap Command:

```
nmap -sT 192.168.1.10
```

4.1.2 Stealth Scan (Half-Open SYN Scan)

SYN scan does not complete the three-way handshake.

Process:

```
Attacker → Target: SYN  
Target → Attacker: SYN/ACK (port open)  
Attacker → Target: RST (abort)
```

Advantages

- **Most common scan type** (Nmap default with privileges)
- Harder to detect than full connect scan
- Faster than TCP connect
- Less likely to be logged by applications
- Requires root/administrator privileges

Nmap Command:

```
nmap -sS 192.168.1.10
```

4.1.3 Inverse TCP Flag Scan

Scans using **abnormal flag combinations** to elicit responses.



RFC 793 Behavior

According to RFC 793:

- **Closed ports** should return RST
- **Open ports** should ignore packets with unusual flag combinations

This behavior enables stealthy port detection.

FIN Scan

Sends packets with only the FIN flag set.

Response Interpretation:

- No response → Open or filtered
- RST → Closed

```
nmap -sF 192.168.1.10
```

NULL Scan

Sends packets with **no flags** set.

```
nmap -sN 192.168.1.10
```

Xmas Scan

Sends packets with **FIN, PSH, and URG** flags set ("lit up like a Christmas tree").

```
nmap -sX 192.168.1.10
```

4.1.4 Xmas Scan

Abnormal packet with multiple flags for fingerprinting.

Characteristics

- Useful for OS fingerprinting

- Bypasses some simple firewalls
- Windows systems may not respond according to RFC 793
- Effective against Unix/Linux systems

4.1.5 TCP Maimon Scan

Sends **FIN/ACK** packets.

Purpose

- Exploits RFC ambiguities in BSD-derived systems
- Named after Uriel Maimon who discovered the technique
- Most modern systems have patched this behavior

```
nmap -sM 192.168.1.10
```

4.1.6 ACK Flag Probe Scan

ACK scan determines firewall rule sets, not port state.

Use Cases

- Mapping firewall rules
- Detecting stateful vs. stateless firewalls
- Identifying filtered ports
- Understanding firewall configurations

Response Analysis:

- RST → Unfiltered
- No response → Filtered

```
nmap -sA 192.168.1.10
```

4.1.7 IDLE/IPID Header Scan

Zombie scan uses a third-party "zombie" host.



How IDLE Scan Works

1. Find a zombie host with predictable IPID sequence
2. Spoof packets from zombie to target
3. Monitor zombie's IPID increment
4. Infer target port state from IPID changes

Advantages:

- Completely spoofed source address
- Target logs show zombie, not attacker
- Ultimate stealth technique

```
nmap -sl zombie_ip target_ip
```

4.1.8 UDP Scan

UDP scanning is challenging due to protocol characteristics.

Challenges

- No handshake mechanism
- Slow and unreliable
- High rate of false positives
- Often rate-limited by OS and firewalls

Response Interpretation

No Response

Port is open or filtered (ambiguous)

ICMP Port Unreachable

Port is closed

UDP Service Banner

Port is open with responding service

```
nmap -sU 192.168.1.10
```

4.1.9 SCTP INIT Scan

SCTP (Stream Control Transmission Protocol) equivalent of SYN scan.

Characteristics

- Used for SCTP-based services (SS7, SIGTRAN)
- Less commonly filtered than TCP
- Stealth alternative for SCTP networks

```
nmap -sY 192.168.1.10
```

4.1.10 SCTP COOKIE ECHO Scan

Stealth SCTP scanning technique.

Advantages

- Does not complete SCTP association
- Bypasses some firewall rules
- Effective on SCTP-aware networks

```
nmap -sZ 192.168.1.10
```

4.1.11 SSDP and List Scan

SSDP (Simple Service Discovery Protocol) for IoT enumeration.

Applications

- IoT device discovery
- UPnP enumeration
- Smart home device identification
- Network printer discovery

List Scan:

Simply lists targets without scanning (DNS resolution only)

```
nmap -sL 192.168.1.0/24
```

4.1.12 IPv6 Scan

IPv6 scanning presents unique challenges.

Characteristics

- Massive address space makes brute-force scanning impractical
- Requires knowledge of SLAAC patterns
- Often less filtered than IPv4
- Many networks have weak IPv6 security

Tools

Nmap

```
nmap -6 fe80::1
```

THC-IPv6

Specialized IPv6 attack toolkit

IPv6 Considerations:

- Slower than IPv4 scanning
- Requires understanding of addressing schemes
- Link-local, unique local, and global addresses

4.2 Port Scanning with AI

AI optimizes port scanning efficiency and accuracy.

AI Contributions

Adaptive Timeout Tuning

Dynamically adjusting timeouts based on network conditions

Predicting Filtering Behavior

Inferring firewall rules from response patterns

Anomaly-Based Service Classification

Identifying services from unusual banner responses

Automated Summarization

Generating human-readable scan reports

Risk Scoring

Prioritizing vulnerabilities based on exposure and exploitability

Intelligent Retry Logic

Adaptive retry strategies for unreliable networks

4.3 Service Version Discovery

Service versioning identifies specific software versions running on open ports.

Techniques

Banner Grabbing

Connecting to services and capturing welcome banners

Nmap Version Detection

```
nmap -sV 192.168.1.10
```

Protocol-Specific Probing

Sending protocol-specific payloads to elicit version information

Information Revealed

- Application name and version
- Operating system hints
- Service configuration details
- Patch levels and build numbers

4.4 Service Version Discovery with AI

AI enhances version detection capabilities.

AI Applications

Banner Parsing

Extracting version information from complex or obfuscated banners

Obfuscation Detection

Identifying modified or disguised banners

Disguised Service Identification

Recognizing services running on non-standard ports

CVE Correlation

Automatically matching versions to known vulnerabilities

Confidence Scoring

Providing probability scores for uncertain identifications

4.5 Nmap Scan Time Reduction Techniques

Optimization Strategies

Aggressive Timing Template

```
nmap -T5 192.168.1.10
```

Fastest timing (may trigger IDS)

Scan Top Ports Only

```
nmap --top-ports 100 192.168.1.10
```

Scans most common 100 ports

Skip DNS Resolution

```
nmap -n 192.168.1.10
```

Avoids reverse DNS lookups

Disable Retries

```
nmap --max-retries 0 192.168.1.10
```

No retransmission of probes

Ping Before Port Scan

```
nmap -sn 192.168.1.0/24 > hosts.txt  
nmap -iL hosts.txt
```

Only scan confirmed live hosts

Use Host Lists

Provide explicit IP lists instead of ranges

Parallel Scanning

```
nmap --min-parallelism 100 192.168.1.10
```

Increase parallel probe count

5. OS Discovery (Banner Grabbing / Fingerprinting)

5.1 OS Discovery Techniques

Primary Methods

Banner Grabbing

Collecting service banners (HTTP, SSH, FTP, SMTP, Telnet)

TCP/IP Stack Fingerprinting

Analyzing TCP/IP implementation differences

TTL / Window Size Analysis

Examining default TTL and TCP window sizes

ICMP Signature Analysis

Evaluating ICMP response characteristics

5.2 Ways to Identify Target OS

Fingerprinting Indicators

Default Port Behavior

OS-specific open port patterns

Time-Based Responses

Timing characteristics of replies

TCP Flag Responses

Reactions to unusual flag combinations

Application Headers

HTTP, SSH, and other protocol banners revealing OS

Network Stack Quirks

- Initial window size
- Window scaling
- Timestamp options
- Maximum segment size
- TCP options ordering

5.3 OS Discovery Tools

Tool Overview

Nmap

```
nmap -O 192.168.1.10
```

Most comprehensive OS detection database

UnicornsCan

High-speed, flexible scanner with OS detection

Xprobe2

Active OS fingerprinting tool using ICMP

p0f

Passive OS fingerprinting (no active probing)

Ettercap

Passive fingerprinting in MITM scenarios

5.4 Nmap Script Engine for OS Discovery

NSE scripts provide targeted OS and service fingerprinting.

Useful Scripts

http-useragent

```
nmap --script http-useragent 192.168.1.10
```

Extracts HTTP server information

smb-os-discovery

```
nmap --script smb-os-discovery 192.168.1.10
```

Detects OS via SMB protocol

ssh-hostkey

```
nmap --script ssh-hostkey 192.168.1.10
```

Reveals SSH key fingerprints and OS hints

Additional Scripts:

- `banner` — Generic banner grabbing
- `ftp-anon` — FTP version and anonymous access
- `telnet-encryption` — Telnet capabilities

5.5 IPv6 OS Fingerprinting

IPv6 OS detection uses unique characteristics.

IPv6 Fingerprinting Methods

Flow Label Field

OS-specific flow label generation patterns

Hop Limit Values

Default hop limit differs by OS (similar to IPv4 TTL)

ICMPv6 Type Responses

Response types and error messages

Extension Header Ordering

OS-specific header ordering preferences

5.6 OS Discovery with AI

AI significantly improves OS fingerprinting accuracy.

AI Enhancements

Signature Pattern Matching

Machine learning models trained on OS signatures

Anomaly Recognition

Identifying modified or hardened systems

Containerized OS Detection

Distinguishing containers from host OS

Predictive OS Identification

Inferring OS from incomplete or ambiguous data

Version Prediction

Estimating patch levels and build versions

5.7 AI-Automated Network Scanning Script

Custom scripts leverage AI for comprehensive scanning.

Script Capabilities

Automated Module Execution

Run Nmap, Hping, and other tools sequentially

Response Parsing

AI-driven interpretation of scan results

Network Mapping

Automated topology visualization

OS/Service Identification

Machine learning classification

Structured Output

JSON, XML, or database export

Vulnerability Correlation

Automatic CVE lookup and risk assessment

Example Workflow

1. **Discovery:** Host discovery across network range
 2. **Enumeration:** Port and service scanning of live hosts
 3. **Fingerprinting:** OS detection and version identification
 4. **Analysis:** AI-driven classification and risk scoring
 5. **Reporting:** Automated report generation with recommendations
-

6. Scanning Beyond IDS and Firewall



Evasion Techniques

These methods help bypass intrusion detection systems (IDS), intrusion prevention systems (IPS), and firewalls by obfuscating or manipulating scanning traffic.

6.1 Packet Fragmentation

Fragmentation splits packets into smaller pieces.

Purpose

- Bypasses simple packet inspection
- Evades signature-based detection
- Fragments payload across multiple packets
- Exploits reassembly weaknesses

Nmap Command:

```
nmap -f 192.168.1.10
```

6.2 Source Routing

Source routing specifies the path packets take.

Characteristics

- Rarely allowed in modern networks
- Can bypass firewall rules
- Mostly blocked due to security concerns
- Useful in legacy environments

Nmap Command:

```
nmap --ip-options "L 192.168.1.1 192.168.1.10" target
```

6.3 Source Port Manipulation

Spoofing source port to appear as trusted traffic.

Common Trusted Ports

Port 53 (DNS)

Often allowed through firewalls for DNS responses

Port 20 (FTP-DATA)

FTP active mode data connections

Port 80 (HTTP)

Web traffic source port

Port 443 (HTTPS)

Secure web traffic

Nmap Command:

```
nmap --source-port 53 192.168.1.10
```

6.4 IP Address Decoy

Decoy scanning includes fake source IP addresses.

Technique

- Generates packets from multiple fake IPs
- Makes it difficult to identify the real attacker
- Target sees scans from many sources
- Overwhelms logging and analysis

Nmap Command:

```
nmap -D RND:10 192.168.1.10  
nmap -D decoy1,decoy2,ME,decoy3 192.168.1.10
```

6.5 IP Address Spoofing

Modifying the source IP address in packets.

Use Cases

IDLE Scans

Using zombie hosts for scanning

Amplification Attacks

DDoS reflection techniques

Bypassing IP-Based ACLs

Evading IP whitelist/blacklist

Limitations

- Cannot receive replies directly
- Requires packet sniffing or IDLE technique
- Ineffective against stateful firewalls

6.6 MAC Address Spoofing

Changing NIC MAC address to avoid tracking.

Applications

- Bypass MAC filtering

- Impersonate trusted devices
- Avoid device tracking
- Reset bandwidth limits

Nmap Command:

```
nmap --spoof-mac Cisco 192.168.1.10  
nmap --spoof-mac 0 192.168.1.10 # Random MAC
```

Manual Spoofing (Linux):

```
ifconfig eth0 down  
ifconfig eth0 hw ether 00:11:22:33:44:55  
ifconfig eth0 up
```

6.7 Creating Custom Packets

Custom packet crafting for advanced evasion.

Tools

Hping3

Command-line packet crafter

Scapy

Python-based interactive packet manipulation

```
from scapy.all import *  
packet = IP(dst="192.168.1.10")/TCP(dport=80,flags="S")  
send(packet)
```

Nemesis

Command-line packet injection tool

6.8 Randomizing Host Order / Bad Checksums

Randomization

Randomize Host Order

Scan targets in random sequence to avoid pattern detection

```
nmap --randomize-hosts 192.168.1.0/24
```

Bad Checksums

Send packets with invalid checksums to test firewall behavior

```
nmap --badsum 192.168.1.10
```

6.9 Proxy Servers

Proxies forward traffic to hide origin.

Types

HTTP Proxies

Web traffic proxying

SOCKS Proxies

Protocol-agnostic proxying (SOCKS4/SOCKS5)

Anonymous Proxies

Remove identifying information from requests

Transparent Proxies

Intercepting proxies (not anonymous)

6.10 Proxy Chaining

Chaining multiple proxies for increased anonymity.

Benefits

- Each proxy only knows previous and next hop
- Extremely difficult to trace back to origin
- Can combine different proxy types
- Geographic obfuscation

Example Chain:

Attacker → SOCKS5 Proxy → HTTP Proxy → TOR Exit Node → Target

6.11 Proxy Tools

Popular Tools

ProxyChains

```
proxychains nmap 192.168.1.10
```

Force any application through proxy chains

Burp Suite

Web application proxy and security testing

TOR (The Onion Router)

Anonymous communication network

Privoxy

Non-caching web proxy with advanced filtering

Shadowsocks

Secure SOCKS5 proxy

6.12 Anonymizers

Anonymizers hide your IP through obfuscation networks.

Services

- VPN services
- Web-based anonymizers
- TOR network
- I2P (Invisible Internet Project)
- Freenet

6.13 Censorship Circumvention Tools

Tools to bypass censorship and filtering:

VPN (Virtual Private Network)

Encrypted tunnel to bypass restrictions

TOR

Onion routing for anonymous browsing

Psiphon

Censorship circumvention tool

Shadowsocks

Lightweight SOCKS5 proxy

V2Ray

Advanced proxy platform

Lantern

P2P censorship circumvention

7. Network Scanning Countermeasures



Defensive Strategy

Organizations must implement multiple layers of defense to detect, prevent, and respond to network scanning activities.

7.1 Ping Sweep Countermeasures

Defensive Measures

Block ICMP Types

Filter ICMP Echo Request (Type 8) and Echo Reply (Type 0)

```
# iptables example
iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
```

Rate Limiting

Limit ICMP response rate to slow down scanning

Network Intrusion Detection

Deploy IDS to detect ping sweep patterns

Hide Non-Public Systems

Place internal systems behind NAT or firewall

7.2 Port Scanning Countermeasures

Protection Strategies

Firewall Rules

Block unnecessary ports and services

SYN Cookies

Protect against SYN flood during scans

Port Knocking

Require specific sequence before opening ports

```
# Example: knock ports 1234, 5678, 9012 to open SSH
```

Dynamic Port Filtering

Temporarily block IPs showing scanning behavior

Service Minimization

Disable unnecessary services and close unused ports

Network Segmentation

Isolate critical systems from scanning attempts

7.3 Banner Grabbing Countermeasures

Mitigation Techniques

Disable Verbose Banners

Remove version information from service banners

Generic Headers

Replace specific banners with generic responses

```
# Nginx example
```

```
server_tokens off;
```

```
# Apache example  
ServerTokens Prod  
ServerSignature Off
```

Banner Randomization

Randomize or falsify banner information

Reverse Proxies

Use reverse proxies to mask backend services

7.4 IP Spoofing Detection

Detection Methods

Packet Filtering

Ingress/egress filtering at network boundaries

Reverse Path Forwarding (RPF)

Verify source IP against routing table

TTL Analysis

Detect inconsistent TTL values

OS Fingerprinting Mismatch

Identify when OS fingerprint doesn't match claimed IP

SYN Cookie Validation

Validate return path for TCP handshakes

7.5 IP Spoofing Countermeasures

Prevention Measures

Anti-Spoof ACLs

Access control lists rejecting spoofed traffic

```
# Block RFC 1918 private addresses from WAN  
iptables -A INPUT -i eth0 -s 10.0.0.0/8 -j DROP
```

```
iptables -A INPUT -i eth0 -s 172.16.0.0/12 -j DROP
iptables -A INPUT -i eth0 -s 192.168.0.0/16 -j DROP
```

Ingress/Egress Filtering

BCP 38 and BCP 84 best practices

Disable IP Source Routing

Reject packets with source routing options

```
# Linux kernel parameter
sysctl -w net.ipv4.conf.all.accept_source_route=0
```

Stateful Inspection

Stateful firewalls tracking connection state

7.6 Scanning Detection Tools

Detection and Prevention Systems

Snort

Open-source network intrusion detection system

```
# Example Snort rule for port scan detection
alert tcp any any → $HOME_NET any (flags: S; detection_filter: track by_src, count 20, seconds 60; msg:"Possible port scan detected";)
```

Suricata

High-performance IDS/IPS engine

Zeek (formerly Bro)

Network analysis framework

OSSEC

Host-based intrusion detection system

SIEM Correlation

Security Information and Event Management platforms

Firewall Logs

Regular analysis of firewall and IDS logs

Additional Tools:

- **PortSentry** — Port scan detection and response
- **Fail2Ban** — Automated ban based on log patterns
- **PSAD** — Port Scan Attack Detector



Module Summary

This module covered comprehensive network scanning techniques, from basic host discovery using ICMP, TCP, and UDP methods to advanced port scanning techniques including stealth scans, fragmentation, and firewall evasion.

Key concepts include:

- TCP/IP communication fundamentals and flag manipulation
- Host discovery techniques (ARP, ICMP, TCP, UDP)
- Port scanning methods (SYN, Connect, FIN, NULL, Xmas, IDLE)
- Service and OS fingerprinting
- Evasion techniques (fragmentation, spoofing, decoys, proxies)
- Defensive countermeasures and detection methods

Network scanning is a critical phase that bridges passive reconnaissance and active exploitation, requiring both technical skill and strategic thinking to execute effectively while avoiding detection.